Republic of Tunisia
The Ministry of Higher Education
and Scientific Research

The University of Carthage

The Higher Institute of Information and
Communication Technologies

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the
BACHELOR DEGREE IN COMPUTER SCIENCE

Field of Study : Software Engineering and Information Systems

---

Design and Implementation of a Network Intrusion Detection System

---

By

AMIRA BOUAFIF & HAJER JEMAA

Conducted within ITXTREE

Publicly defended on June 6, 2026 in front of the jury members:

Chairman:	Name SURNAME, University Relations Leader, IBM
Reporter:	Name SURNAME, Teacher, ISTIC
Examiner:	Name SURNAME, Teacher, ISTIC
Professional Supervisor:	Firas ABBESSI, Engineer, ITXTREE
Academic Supervisor:	Wassim ABBESSI, Teacher, ISTIC

Academic Year: 2025-2026

Republic of Tunisia
The Ministry of Higher Education
and Scientific Research

The University of Carthage

The Higher Institute of Information and
Communication Technologies

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the
BACHELOR DEGREE IN COMPUTER SCIENCE

Field of Study : Software Engineering and Information Systems

Design and Implementation of a Network Intrusion Detection System

By

AMIRA BOUAFIF & HAJER JEMAA

Conducted within ITXTREE

Authorization of graduation project report submission:

Professional Supervisor:

Issued on :

Signature:

Academic Supervisor:

Issued on :

Signature:

Dedications

I dedicate this End-of-Study project

To my dear parents

Who have supported and encouraged me every step of the way. This is for you, in honor of all the sacrifices you have made for me. I hope to have made you both proud and happy.

To my dear sisters and brothers

Who have been a great support and a source of joy in my life. I am grateful for your love and encouragement throughout this journey.

To all my family and friends

Thank you all for your support and encouragement throughout all this time. I am truly blessed to have you in my life.

Amira BOUAFIF

Dedications

I dedicate this End-of-Study project

To my dear parents

To my dear sisters

To all my family and friends

Hajer JEMAA

Acknowledgments

We would like to extend our sincere gratitude to all those who have supported us throughout the course of this End-of-Study project. Their guidance, encouragement, and support have been invaluable throughout this journey.

First and foremost, we would like to thank our academic supervisor, Professor Wassim ABBESSI, for his guidance and insights, which have been a great help in the successful completion of this project.

Our gratitude also go to ITXTREE and, in particular, our professional supervisor, Mr. Firas ABBESSI, for providing us with the opportunity to work on this project and for their professional mentorship.

We also extend our deep appreciation to the professors at the Higher Institute of Information and Communication Technologies for their invaluable support and teachings, which have greatly contributed to our academic journey.

Finally, we thank everyone who has, in one way or another, helped in making this project a success.

Contents

List of Figures

List of Tables

General Introduction

Chapter 1

Project Context

1.1 Introduction

This chapter serves as an introduction to the project. We begin with a presentation of the company hosting the internship, a study of existing network intrusion detection systems. Then, we present the problem and the proposed solution. Finally, we end the chapter with a conclusion.

1.2 Presentation of the company

ITXTREE is an early-stage technology startup based in Sidi Rzig, Tunisia, focused on building a solid foundation for future digital solutions it is built on the core values of integrity, quality, curiosity and adaptability. Figure 1.1 illustrates the company logo.

Figure 1.1 – ITXTREE Logo [1]

1.3 Study of the existing

1.4 Criticism of the existing

1.5 Proposed solution

Our goal is developing a network intrusion detection system that addresses the identified shortcomings through the following features:

1.6 Project Pipeline

A project pipeline is a structured sequence of stages that outlines the workflow and processes involved in the development and deployment of a project. It serves as a roadmap, guiding the team through various phases, from initial planning to final delivery.

Figure 1.2 illustrates the project pipeline for our network intrusion detection system.

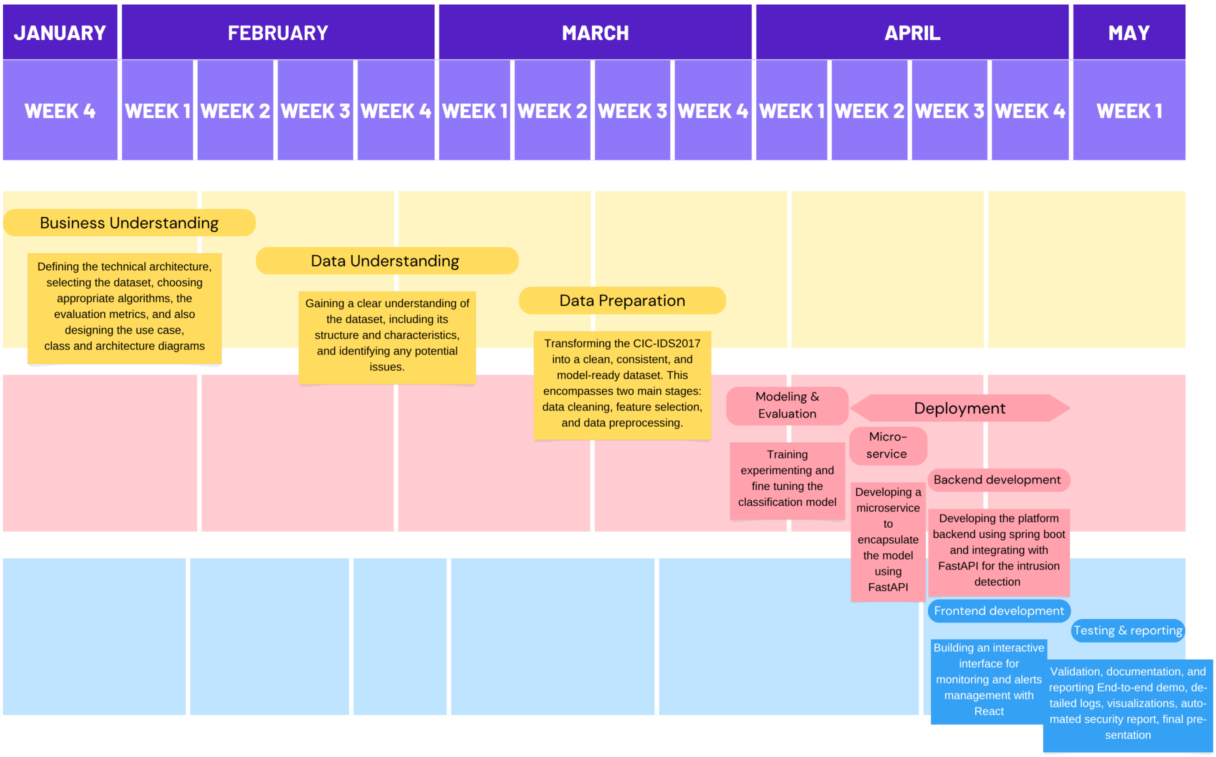

FIGURE 1.2 – Project Pipeline

1.7 Methodologies

In this section, we present the methodologies that we use in this project, including the machine learning methodology CRISP-DM and the software development methodology UML.

1.7.1 CRISP-DM

The CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology is a widely used framework for structuring machine learning projects. Figure 1.3 illustrates the six phases of this methodology and their relationships.

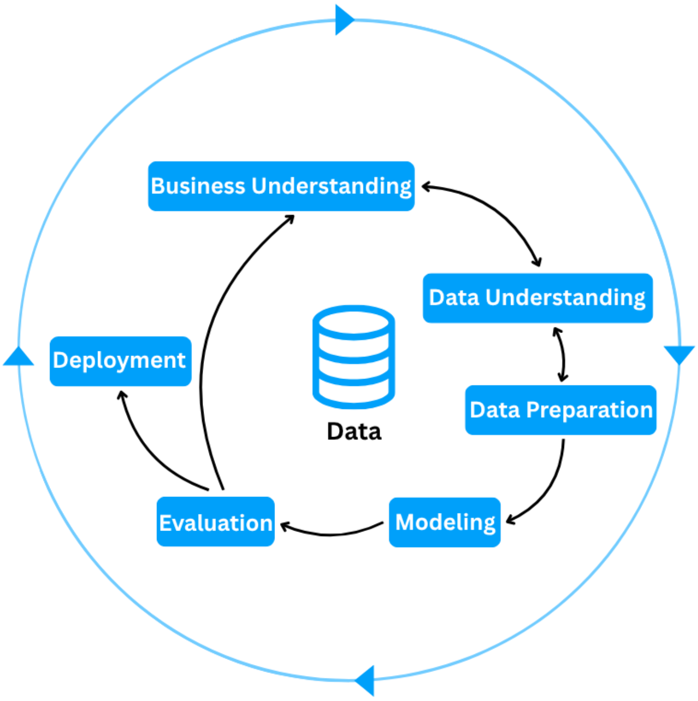

FIGURE 1.3 – CRISP-DM Methodology

1.7.2 UML

For the modeling of the software system, we use UML (Unified Modeling Language), a standardized modeling language used to specify, visualize, construct, and document the artifacts of software systems through diagrams such as use case, class, and sequence diagrams.

FIGURE 1.4 – UML Logo [2]

1.8 Conclusion

In this chapter we introduce the host company ITXTREE, study existing network intrusion detection systems, discuss their shortcomings and finally present a solution. In the next chapter we specify the system requirements.

Chapter 2

Requirements Specification

2.1 Introduction

Throughout this chapter, we specify both the functional and non-functional requirements, identify the system actors, and present the global use case diagram. By defining these elements, we aim to establish a clear and shared understanding of the system's expected behavior and constraints, providing a solid foundation for the subsequent design and implementation phases.

2.2 Functional Requirements

Functional requirements define *what* the system should perform. The main functionalities of our system are as follows:

- Authenticate;
- View Dashboard;
- View Statistics summary;
- Trigger Network Analysis;
- View Alerts List;
- View Reports;

2.3 Non-Functional Requirements

Non-functional requirements define *how* the system should perform its functions. They describe the quality attributes and constraints of the system:

Performance: The system must process network traffic and generate analysis results with minimal latency, ensuring fast detection of potential intrusions.

Security: The system must ensure secure authentication and enforce role-based access control, preventing unauthorized actions and protecting system integrity, while supporting non-repudiation to guarantee accountability and traceability of all operations.

Reliability: The system must perform its functions consistently, handle errors gracefully, and ensure that failures are detected and managed without disrupting overall operation.

Maintainability: The system should be easy to maintain, allowing for efficient debugging, updates, and modifications to existing components.

Extensibility: The system should support the addition of new features, such as new detection models or data sources, without requiring major redesign.

Availability: The system must be accessible and operational whenever needed, minimizing downtime and ensuring continuous monitoring of network activity.

Usability: The system should provide an intuitive interface, enabling users to efficiently navigate and interact with its functionalities.

2.4 Actors Identification

Actors are external entities, users, organizations, or external systems that interact with the system to achieve one or more functional requirements, with each actor representing a specific role.

In the context of our project, we identify two primary actors illustrated in Figure 2.1

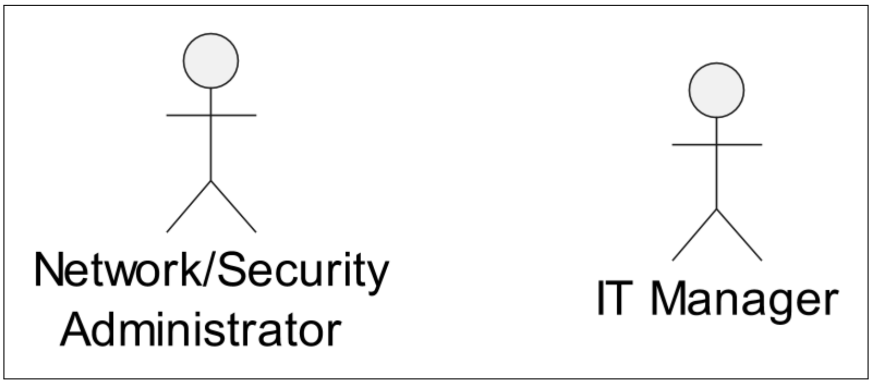

FIGURE 2.1 – Actors

Table 2.1 presents the roles of each actor in our system.

TABLE 2.1 – Roles of Actors

Actor	**Role**
Network/Security Administrator	- Authenticate - View Dashboard - View Statistics Summary - Trigger Network Analysis - View Alerts List
It Manager	- Authenticate - View Dashboard - View Statistics Summary - View Reports

2.5 Use Case Diagram

A use case diagram is a visual representation of *how* users interact with the system and *what* a system does through different use cases. Figure 2.2 illustrates that diagram.

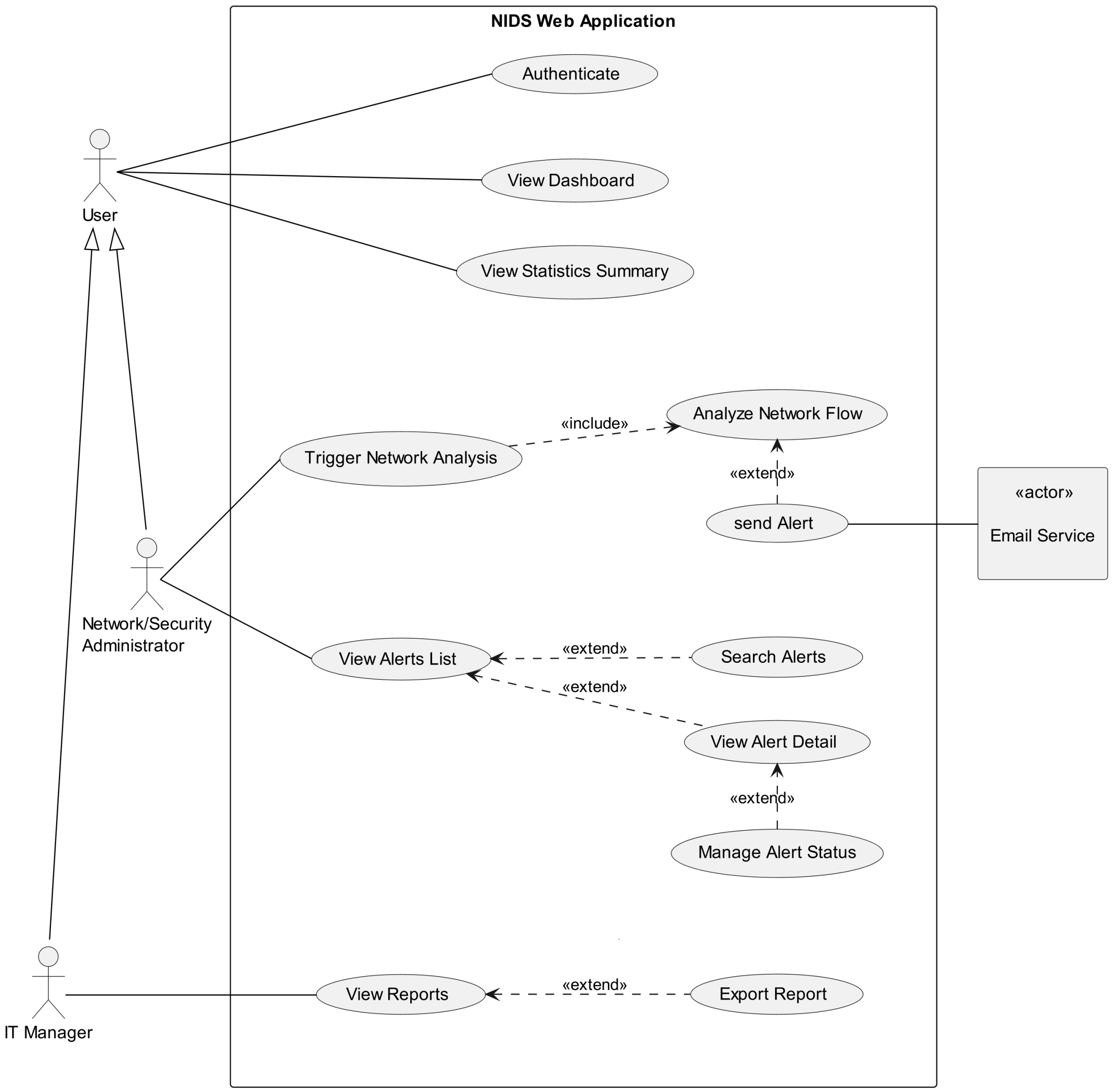

Figure 2.2 – Use Case Diagram

2.6 Conclusion

In this chapter, we outline the functional and non-functional requirements of our system, identify the primary actors, and represent the way they interact with the system and the system's functionality via a use case diagram. In the next chapter, we present the conceptual design of the platform.

Chapter 3

Conceptual Design

3.1 Introduction

3.2 Sequence Diagrams

3.3 Class Diagram

3.4 Deployment Diagram

3.5 Conclusion

Chapter 4

Data Collection and Preparation

4.1 Introduction

4.2 Data Collection

4.3 Data Understanding

4.4 Data Preparation

4.5 Data Augmentation

4.6 Conclusion

Chapter 5

Modeling

5.1 Introduction

5.2 State of the Art

5.3 Adopted Approach

5.4 Training and Hyperparameter Optimization

5.5 Conclusion

Chapter 6

Realization

6.1 Introduction

6.2 Development Tools and Technologies

6.3 Performance Analysis and Evaluation Metrics

6.4 User Interfaces

6.5 Integration

6.6 Conclusion

General Conclusion

Webography

[1] ITXTREE – Logo. https://tn.linkedin.com/company/itxtree, Accessed on: April 2, 2026.

[2] Unified Modeling Language – logo. https://en.wikipedia.org/wiki/Unified_Modeling_Language. Accessed on: April 2, 2026.